Task 2 Numerical Methods

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PROBLEM 1

No Solution

In this case, the equations will be inconsistent, meaning that they can't all be true at the same time.

$$x + y + z = 1$$
$$2x + 2y + 2z = 2$$
$$3x + 3y + 3z = 5$$

One Solution Only

In this case, the equations will be independent and intersect at exactly one point in $\mathbb{R}^3$.

$$x + y + z = 1$$
$$x - y + z = 3$$
$$2x + 3y - z = 1$$

Solving these equations yields the unique solution $(x, y, z) = (1, -1, 1)$.

Infinitely Many Solutions

In this case, the equations are dependent and represent planes that intersect along a line or coincide.

$$x + y + z = 1$$
$$2x + 2y + 2z = 2$$
$$3x + 3y + 3z = 3$$

All three equations are multiples of each other and therefore represent the same plane in $\mathbb{R}^3$. Any point on that plane is a solution, so there are infinitely many solutions.

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PROBLEM 2

4.2.30

Crout's method allows me to compute L and U as follows:

For $i = 1, 2, 3, \ldots, n$ (where n is the dimension of the square matrix A):

1. $u_{ii} = 1$

2. $l_{ij} = a_{ij} - \sum_{k=1}^{i-1} l_{ik} u_{kj}$ (for $j = i, i+1, \ldots, n$)

3. $u_{ij} = \frac{1}{l_{ii}} \left(a_{ij} - \sum_{k=1}^{i-1} l_{ik} u_{kj} \right)$ (for $j = i+1, i+2, \ldots, n$)

Let's start with the given matrix A :

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & -1 & 3 \\ 1 & 3 & 0 \end{bmatrix}$$

I then initialize L and U to zero matrices of the same dimensions as A . and calculate L and U:

i=1:

$\cdot\, u_{11} = 1$

$\cdot\, l_{11} = 3 - 0 = 3$

$\cdot\, l_{21} = 0 - 0 = 0$

$\cdot\, l_{31} = 1 - 0 = 1$

$\cdot\, u_{12} = (0 - 0)/3 = 0$

$\cdot\, u_{13} = (1 - 0)/3 = 1/3$

i=2 :

$\cdot\, u_{22} = 1$

$\cdot\, l_{22} = -1 - 0 = -1$

$\cdot\, l_{32} = 3 - 0 = 3$

$\cdot\, u_{23} = (3 - 0)/(-1) = -3$

i=3 :

$\cdot\, u_{33} = 1$

$\cdot\, l_{33} = 0 - (1 \times -3) = 3$

After this i get:

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 1 & 3 & 3 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 & 1/3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

4.2.31

The given matrix A is:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$$

and the L matrix is defined as follows:

$$L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}$$

I will use the equations $A = LL^T$ to find l_{11}, l_{21}, and l_{22}:

$$l_{11}^2 = 1 \Rightarrow l_{11} = 1$$

$$l_{21} \times l_{11} = 2 \Rightarrow l_{21} = 2$$

$$l_{21}^2 + l_{22}^2 = 5 \Rightarrow l_{22} = \sqrt{5-4} \Rightarrow l_{22} = \sqrt{1} \Rightarrow l_{22} = 1$$

The L matrix then becomes:

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

Now $A = LL^T$:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$$

So, the Cholesky decomposition of A into LL^T results in L being the 2×2 matrix $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

4.2.32

To find the LL^T-factorization of A, where L is a lower triangular matrix with positive diagonal elements, I set up L as:

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

We then have $A = LL^T$. I can now write the equations as follows:

$$l_{11}^2 = 4 \Rightarrow l_{11} = 2$$

$$l_{21} \times l_{11} = \frac{1}{2} \Rightarrow l_{21} = \frac{1}{4}$$

$$l_{31} \times l_{11} = 1 \Rightarrow l_{31} = \frac{1}{2}$$

$$l_{21}^2 + l_{22}^2 = \frac{17}{16} \Rightarrow l_{22} = \sqrt{\frac{17}{16} - \frac{1}{16}} \Rightarrow l_{22} = \sqrt{\frac{16}{16}} \Rightarrow l_{22} = 1$$

$$l_{31} \times l_{21} + l_{32} \times l_{22} = \frac{1}{4} \Rightarrow l_{32} = \frac{1}{4} - \frac{1}{8} \Rightarrow l_{32} = \frac{1}{8}$$

$$l_{31}^2 + l_{32}^2 + l_{33}^2 = \frac{33}{64} \Rightarrow l_{33} = \sqrt{\frac{33}{64} - \frac{4}{64} - \frac{1}{64}} \Rightarrow l_{33} = \sqrt{\frac{28}{64}} \Rightarrow l_{33} = \frac{1}{2}$$

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After solving these equations i get the matrix L:

$$L = \begin{bmatrix} 2 & 0 & 0 \\ \frac{1}{4} & 1 & 0 \\ \frac{1}{2} & \frac{1}{8} & \frac{1}{2} \end{bmatrix}$$

So, the LL^T-factorization of A with L having positive diagonal elements results in the above L matrix.

4.4.16

To find the condition number of a matrix A using the infinity matrix norm, I can use the formula:

$$\text{cond}(A) = \|A\|_\infty \times \left\|A^{-1}\right\|_\infty$$

Here, A is the given matrix

$$A = \begin{bmatrix} 7 & 8 \\ 9 & 10 \end{bmatrix}$$

First, I will find $\|A\|_\infty$: The infinity norm of A is the maximum row sum:

$$\|A\|_\infty = \max(7+8, 9+10) = \max(15, 19) = 19$$

Next, we find A^{-1} using Cramer's Rule. The determinant of A, denoted $\det(A)$, is:

$$\det(A) = (7 \times 10) - (8 \times 9) = 70 - 72 = -2$$

Using Cramer's Rule, the inverse A^{-1} can be found as:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 10 & -8 \\ -9 & 7 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ 4.5 & -3.5 \end{bmatrix}$$

Now I will find $\left\|A^{-1}\right\|_\infty$: The infinity norm of A^{-1} is the maximum row sum:

$$\left\|A^{-1}\right\|_\infty = \max(-5+4, 4.5+3.5) = \max(-1, 8) = 8$$

Finally, the condition number is:

$$\text{cond}(A) = 19 \times 8 = 152$$

So the condition number of the matrix A using the infinity norm is 152 .

Problem 4

The Neumann Series provides a way to find the inverse of a matrix A as:

$$A^{-1} = I + (I - A) + (I - A)^2 + (I - A)^3 + \ldots$$

The series will converge if $\|I - A\| < 1$.
To find A, I subtract B from the identity matrix I :

$$A = I - B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 0.5 & -0.2 & -0.2 \\ 0.1 & 0.7 & -e \\ 0.1 & -0.3 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.2 & 0.2 \\ -0.1 & 0.3 & e \\ -0.1 & 0.3 & 0.5 \end{pmatrix}$$

To determine for what values of e the Neumann series converges, I will find the infinity matrix norm $\|A\|_\infty$ and determine when $\|A\|_\infty < 1$.

The infinity matrix norm of a matrix A is defined as

$$\|A\|_\infty = \max_i \left(\sum_j |a_{ij}| \right)$$

For matrix A, the sum of the absolute values of the rows are:

$$0.5 + 0.2 + 0.2 = 0.9$$

$$0.1 + 0.3 + |e| = 0.4 + |e|$$

$$0.1 + 0.3 + 0.5 = 0.9$$

The largest sum here is $\max(0.9, 0.9, 0.4 + |e|)$. For the Neumann series to converge, this value must be smaller than 1 . Therefore,

$$\max(0.9, 0.9, 0.4 + |e|) < 1$$

This inequality is satisfied for $|e| < 0.6$. So, for e in the interval $-0.6 < e < 0.6$, the Neumann Series will converge to the inverse of B.

Problem 5

The condition number $\kappa(A)$ of a matrix A with respect to a given norm is defined as:

$$\kappa(A) = \|A\| \times \left\|A^{-1}\right\|$$

Firstly, I will find A^{-1}. Since A is a diagonal matrix, A^{-1} is simply the diagonal matrix where each diagonal entry a_{ii} is replaced by $1/a_{ii}$.

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{1} & 0 \\ 0 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{1} & -\frac{e}{1} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -e \\ 0 & 0 & 1 \end{pmatrix}$$

Now, we find the infinity norms $\|A\|_\infty$ and $\left\|A^{-1}\right\|_\infty$. The infinity norm of a matrix is given by the maximum absolute row sum.

For A, each row sum is $|1|, |1 + e|, |1|$. Therefore, $\|A\|_\infty = \max(1, 1 + |e|, 1) = 1 + |e|$.

Similarly, for A^{-1}, each row sum is $|1|, |1 + |e||, |1|$. Therefore, $\left\|A^{-1}\right\|_\infty = \max(1, 1+ |e|, 1) = 1 + |e|$. So, the condition number $\kappa(A)$ as a function of e is:

$$\kappa(A) = (1 + |e|) \times (1 + |e|) = (1 + |e|)^2$$